

Topic: Cardiovascular Disease Risk Assessment Data Visualization Report using Shinny

Introduction

Cardiovascular diseases (CVDs) are the leading cause of mortality worldwide, accounting for approximately 18 million deaths annually. They are responsible for at least 38% of the 18 million premature deaths attributed to noncommunicable diseases each year (WHO).

There are modifiable and non-modifiable factors. Among these, modifiable risk factors are particularly important because they can be targeted through disease prevention strategies and public health interventions. These include unhealthy dietary habits, physical inactivity, tobacco use, and harmful alcohol use. In addition, several metabolic and physiological risk factors, such as hypertension, diabetes mellitus, abdominal obesity, and dyslipidemia, play a significant role in the development of cardiovascular disease. Collectively, these risk factors account for approximately 90% of the risk of a first myocardial infarction.

Understanding the epidemiology and major risk factors for cardiovascular disease matters both for individuals and for public health. For individuals, this knowledge makes it possible to reduce personal risk through practical lifestyle changes. For public health systems, it provides the evidence needed to design and implement prevention, health-promotion, and risk-reduction strategies that lower the overall burden of cardiovascular disease.

Data Cleaning and Visualization

In this project, I developed a simple yet powerful data visualization dashboard that enables users to interactively explore cardiovascular disease (CVD) risk factors by identifying relationships between key variables and the likelihood of CVD. Ultimately, the goal is to empower decision-makers with real-time insights derived from patient data to guide clinical decision-making and public health interventions.

To begin the project, I obtained the dataset from Kaggle. The original dataset contains 22 categorical and numerical variables representing major cardiovascular risk factors, including diabetes, smoking status, physical activity level, hypertension, and total cholesterol, among others. The initial phase of the project involved data cleaning and preprocessing. Records with missing values and duplicate entries were removed to ensure data quality. The risk level variable was converted into both a factor and a numeric format to facilitate visualization and correlation analysis.

Once the data was cleaned, exploratory analysis and visualizations were conducted. Using ggplot2, bar plots were created to show the distribution of patients by risk level. Box plots illustrated how age, blood pressure, cholesterol, fasting blood sugar and other variables associated with cardiovascular disease risk level. These visualizations provided meaningful, actionable insights about the structure of the data and the severity of risk across groups.

The next phase involved building the Shiny application itself. The app allows users to interactively explore the dataset by selecting variables from a dropdown menu and viewing corresponding box plots of those variables stratified by risk level. The interface is structured with a sidebar for user input and a main panel for dynamic plot rendering. When a user changes the selected variable, the output updates in real time, enabling exploration without needing to write any R code.

Implementation Details

The project was implemented entirely in R using RStudio as the primary development environment. Key packages included `readr` for data import, `dplyr` for data cleaning and manipulation, `ggplot2` and `plotly` for data visualization, and `shiny` and `shinydashboard` for developing the interactive web application.

The cardiovascular disease (CVD) risk assessment dataset was initially imported into R using the `read_csv()` function., cleaned using a sequence of `dplyr` functions, and subsequently exported as a new CSV file for use in the dashboard application. Following data cleaning, exploratory data analysis and visualization were conducted using `ggplot2`. Then the Shiny dashboard was developed using a combination of `fluidPage()`, `sidebarLayout()`, and reactive server functions.

The app was deployed using the `rsconnect` package to shinyapps.io, and the full source code was published to GitHub. This enables others to review the code, reproduce the analysis, and deploy the application independently.

In conclusion, this project demonstrates how open-source digital health tools can be used to transform raw health data into accessible, interactive, and evidence-based visualizations that support public health education, risk communication, and data-driven decision-making in cardiovascular disease prevention.

Links:

- Shiny App: <https://adam-g.shinyapps.io/DHP1/>
- GitHub Repository: [https://github.com/adamghebre1992-sudo/CVD shinny dashboard](https://github.com/adamghebre1992-sudo/CVD_shinny_dashboard)